

Math 421

Wednesday, November 5

Announcements

- Homework 8 Due **today**
- Exam 2 – 11/12 5:45-7:15 pm
 - **Date:** Wednesday, 11/12
 - **Time:** 5:45-7:15 pm
 - **Location:** Birge 145
- Exam Week Drop-In Hours
 - Tuesday 1:30-3:30 pm VV B205
 - Wednesday 12:00-1:00 pm VV B205

Discussion Questions/Activity

1. What would you need to do to prove that $\lim_{x \rightarrow a} f(x)$ does not exist?
 - a) According to the $\epsilon - \delta$ definition of the limit
 - b) According to the sequence definition of the limit
2. Show that $\lim_{x \rightarrow 0} \sin\left(\frac{1}{x}\right)$ does not exist, using 1.b).

Algebraic Limit Laws

Theorem. Let f and g be functions defined on a domain S and let a be the limit of some sequence in S . If the limits $L = \lim_{x \rightarrow a} f(x)$ and $M = \lim_{x \rightarrow a} g(x)$ exist and are finite. Then

- i. $\lim_{x \rightarrow a} kf(x)$ exists and equals kL for all $k \in \mathbb{R}$.
- ii. $\lim_{x \rightarrow a} (f + g)(x)$ exists and equals $L + M$.
- iii. $\lim_{x \rightarrow a} (fg)(x)$ exists and equals LM .
- iv. $\lim_{x \rightarrow a} (f/g)(x)$ exists and equals L/M provided $M \neq 0$ and $g(x) \neq 0$ for $x \in \text{dom}(g)$.

One-Sided & Infinite Limits

- $\lim_{x \rightarrow a^+} f(x) = L$

- $\lim_{x \rightarrow \infty} f(x) = L$

- $\lim_{x \rightarrow a} f(x) = \infty$

Activity

Consider each of the following limits, fill in the following blanks for each:

for each _____ there exists _____ such that

_____ implies _____

1. $\lim_{x \rightarrow a^-} f(x) = \infty$

2. $\lim_{x \rightarrow -\infty} f(x) = -\infty$

Chapter 3: Continuity

Section 17: Continuous Functions

Continuity

Definition. A function $f(x)$ is said to be **continuous at a point** $x_0 \in \text{dom } f$ if $\lim_{x \rightarrow x_0} f(x) = f(x_0)$.

If $f(x)$ is continuous at every $x_0 \in S$, then $f(x)$ is said to be a **continuous on S** . If $S = \text{dom } f$, $f(x)$ is called a **continuous function**.

Example

1. Recall we proved $\lim_{x \rightarrow 1} x^2 + x + 1 = 3$. Is $x^2 + x + 1$ continuous at 1?
2. Recall we proved $\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right) = 0$. Define function $g(x)$ so that $g(x) = x^2 \sin\left(\frac{1}{x}\right)$ for all $x \neq 0$ and $g(x)$ is continuous at all $x \in \mathbb{R}$.

Activity – Discontinuity

Sketch at least three different ways a function can fail to be continuous at a point a .

Activity

Prove, using the $\epsilon - \delta$ definition, the following basic properties of continuous functions

1. If $a \in \mathbb{R}$ and $g(x) = a$ is the constant function, then g is continuous at all $c \in \mathbb{R}$.
2. If $f(x) = x$, then f is continuous at all $c \in \mathbb{R}$.

Example

$$\text{Let } f(x) = \begin{cases} 1, & x \in \mathbb{Q} \\ 0, & x \notin \mathbb{Q} \end{cases}$$

Example – Popcorn Function

$$\text{Let } f(x) = \begin{cases} \frac{1}{q}, & x \in \mathbb{Q} \text{ and } x = \frac{p}{q} \text{ in lowest terms} \\ 0, & x \notin \mathbb{Q} \end{cases}$$

Algebraic Continuity Properties

Theorem 17.3 & 17.4. Assume $f: A \rightarrow \mathbb{R}$ and $g: A \rightarrow \mathbb{R}$ are both continuous at a point $x_0 \in A$. Then

1. $|f(x)|$ is continuous at x_0 ,
2. $kf(x)$ is continuous at x_0 for all $k \in \mathbb{R}$,
3. $f(x) + g(x)$ is continuous at x_0 ,
4. $f(x)g(x)$ is continuous at x_0 , and
5. If $g(x_0) \neq 0$, then $\frac{f(x)}{g(x)}$ is continuous at x_0 .

Example

1. If P is a polynomial, then P is continuous at all $c \in \mathbb{R}$
2. Define $f: [0, \infty) \rightarrow [0, \infty)$ by $f(x) = \sqrt{x}$. Then f is continuous at all $c \in [0, \infty)$.

Composition of Continuous Functions

Theorem 17.5. Suppose $f: A \rightarrow \mathbb{R}$ and $g: B \rightarrow \mathbb{R}$. Assume $\{f(x): x \in A\} \subseteq B$ so that the composition $(g \circ f)(x) = g(f(x))$ is defined on A .

If f is continuous at $x_0 \in A$ and g is continuous at $f(x_0) \in B$, then the $g \circ f$ is continuous at x_0 .